

LOCH GLASCARNOCH, United Kingdom

WMO: 030310

Lat: 57.7250N

Lon: 4.8956W

Elev: 866

StdP: 14.24

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.7	20.8	13.9	11.5	17.0	19.1	14.8	23.3	41.7	40.8	36.6	39.9	1.4	270	0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	71.1	60.5	67.2	58.4	64.2	56.9	62.1	68.6	60.0	65.5	58.2	62.9	10.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.4	78.0	64.6	57.5	72.7	62.6	55.7	68.2	60.7	28.2	68.6	26.8	65.6	25.5	62.9	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.0	28.1	24.8	DB	11.6	76.7	5.4	3.4	7.7	79.1	4.5	81.1	1.5	83.0	-2.5	85.4	(4)
(5)			WB	11.9	66.0	5.4	3.2	8.0	68.3	4.9	70.2	1.9	72.0	-2.0	74.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.5	35.9	36.3	38.4	42.6	46.8	51.7	55.0	54.2	51.2	45.7	40.0	36.0		
	DBStd	8.89	6.23	6.02	5.63	5.27	5.52	4.69	4.37	4.15	4.42	5.03	6.02	7.55		
	HDD50	2514	439	385	361	230	130	34	4	8	37	148	302	436		
	HDD65	7480	904	805	826	673	563	400	310	336	415	598	750	900		
	CDD50	511	0	0	0	6	32	84	160	139	72	15	2	1		
	CDD65	2	0	0	0	0	0	0	1	1	0	0	0	0		
	CDH74	35	0	0	0	0	5	9	13	8	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	10.3	11.4	11.9	12.0	10.4	9.7	9.7	8.7	8.9	9.7	10.2	10.2	10.6		
(15) Precipitation	PrecAvg	61.7	7.8	5.8	5.1	3.8	3.9	3.1	3.1	3.8	4.5	6.5	7.2	7.2		
	PrecMax	76.4	14.3	12.1	13.0	6.5	8.3	4.6	5.1	9.3	8.7	11.4	12.4	13.7		
	PrecMin	40.2	0.6	0.8	0.8	1.8	0.5	1.4	1.4	1.5	1.5	2.0	2.6	0.2		
	PrecStd	9.6	4.0	3.2	2.5	1.2	1.9	0.9	1.1	1.9	2.0	2.6	2.6	3.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	53.1	57.2	64.7	72.9	74.5	75.6	74.3	69.3	60.6	55.9	54.0		
		MCWB	48.6	48.7	47.2	52.7	59.1	61.0	63.3	63.4	59.3	54.6	52.2	50.4		
	2%	DB	49.4	50.5	51.4	58.5	66.4	67.9	71.0	67.7	64.3	57.6	53.1	51.5		
		MCWB	46.3	47.0	46.6	49.8	54.6	57.7	61.6	60.0	57.4	53.4	49.9	48.6		
	5%	DB	47.2	48.1	49.1	54.9	61.4	63.6	66.9	64.2	61.2	55.5	51.0	49.5		
		MCWB	44.2	45.0	45.3	48.1	52.1	55.6	59.5	57.9	56.0	51.9	47.9	46.7		
	10%	DB	44.9	45.4	47.2	51.8	57.2	60.2	63.4	61.6	58.9	53.6	48.9	46.8		
		MCWB	42.3	42.4	43.8	46.4	50.2	53.9	57.5	56.5	54.4	50.0	46.2	44.1		
	0.4%	WB	48.9	49.9	49.5	53.8	60.7	62.6	65.2	64.7	61.1	56.3	52.9	51.2		
		MCDB	51.5	52.4	53.6	60.5	71.2	71.6	72.2	70.2	65.5	59.1	55.3	53.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.8	47.6	47.4	50.9	55.9	59.1	62.5	61.2	58.5	54.2	50.5	49.0		
		MCDB	49.0	50.3	50.7	57.3	64.1	66.5	69.0	66.3	62.6	56.9	52.9	51.3		
	5%	WB	44.6	45.4	45.8	48.9	53.2	56.7	60.3	59.0	56.7	52.4	48.3	46.9		
		MCDB	47.0	48.0	48.9	53.9	60.2	62.0	65.8	63.0	60.5	55.0	50.4	49.2		
	10%	WB	42.4	42.7	44.0	46.9	50.7	54.6	58.2	57.2	54.9	50.6	46.4	44.3		
		MCDB	44.7	45.2	46.9	51.1	56.1	59.3	62.9	60.8	58.4	53.2	48.7	46.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.6	11.6	13.9	15.7	13.7	13.3	12.9	13.2	11.5	10.3	10.5		
		MCDBR	10.8	12.4	14.9	20.4	24.5	20.3	20.3	18.2	17.4	13.6	11.9	12.3		
		MCWBR	9.3	10.1	11.0	12.7	14.4	11.9	12.0	11.5	11.6	10.4	10.3	11.3		
	5% WB	MCDBR	10.4	11.2	13.5	17.9	22.1	18.5	18.2	16.4	15.8	12.4	11.6	12.1		
		MCWBR	9.4	9.9	10.7	11.9	13.7	11.7	11.6	11.5	11.5	10.3	10.6	11.3		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.336	0.356	0.372	0.374	0.367	0.369	0.356	0.347	0.333	0.313	0.304		
	taud		2.157	2.253	2.282	2.291	2.336	2.390	2.397	2.439	2.458	2.394	2.228	2.101		
	Ebn at Noon		161	219	249	265	273	276	273	269	253	220	163	125		
	Edh at Noon		17	24	30	35	36	35	34	30	26	21	16	13		
(44) All-Sky Solar Radiation	RadAvg		115	305	628	1068	1440	1406	1320	1056	730	381	156	80		
	RadStd		16	39	52	78	110	137	126	66	55	41	15	9		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page